Swallowing Safety With Riluzole Oral Film

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BACKGROUND

- Dysphagia is highly prevalent in patients with ALS[1]
- Riluzole is a US Food and Drug Administration–approved treatment with survival benefits in ALS[2]
- Current riluzole administration routes require swallowing a tablet or liquid form that can be physically challenging and burdensome to patients with ALS
- Riluzole oral film (ROF; Exservan™) contains riluzole in a polymer-based film matrix that is applied and dissolved onto the tongue[3]

OBJECTIVE

- To describe the ROF bioavailability and pharmacokinetic data

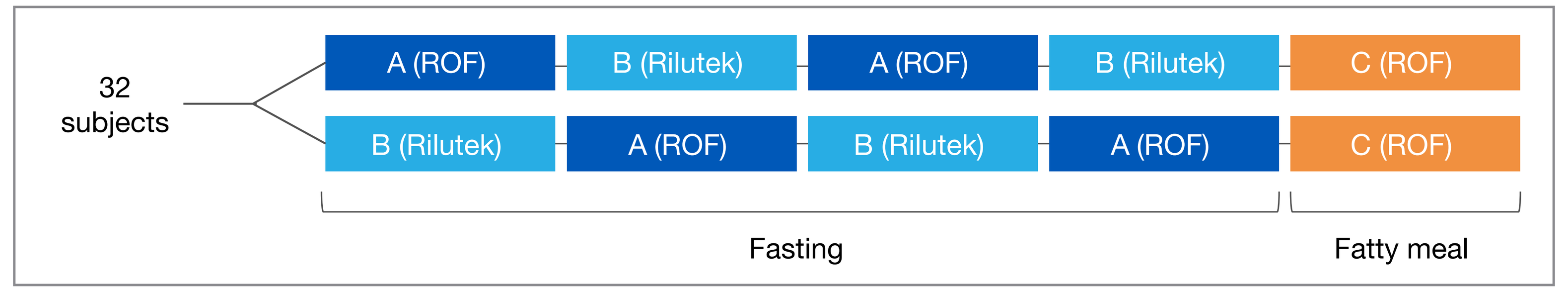
METHODS

2 clinical trials were conducted

Study 162020

- Pivotal bioavailability and pharmacokinetics (PK) study
- Phase 1, open-label, randomized, single-dose, 5-period, replicate crossover, comparative bioavailability study in 32 healthy volunteers (**Figure 1**)
- Compared riluzole 50 mg ROF and Rilutek® 50 mg tablets under fasting conditions with evaluation of food effect
 - Treatment A: ROF 50 mg (fasting)
 - Treatment B: Rilutek 50 mg (fasting)
 - Treatment C: ROF 50 mg (fatty meal)

Figure 1. Study 162020 Study Design


32 subjects
A (ROF)
B (Rilutek)
A (ROF)
B (Rilutek)
C (ROF)
B (Rilutek)
A (ROF)
B (Rilutek)
A (ROF)
C (ROF)
Fasting
Fatty meal

Study 17MOIR-0012

- Phase 2, single-center study to evaluate the effect of ROF on swallowing safety in individuals with ALS
- This videofluoroscopic swallowing study was conducted with 9 patients with ALS before and after receiving a single dose of 50 mg ROF
- The primary outcome was change on penetration-aspiration scale (PAS) scores from pre- to post-dose
- Inclusion Criteria
 - 18-80 years; diagnosis of probable/definite ALS (revised El-Escorial criteria); consuming oral diet (Functional Oral Intake Scale ≥3); no known allergy to barium sulfate, riluzole, or inactive ingredients in ROF
- Videofluoroscopic Swallowing Study
 - The Videofluoroscopic Swallowing Study (VFSS) is considered the gold standard for studying swallowing and its dysfunction[4]
 - VFSS permits the visualization of bolus flow in relation to structural movement throughout the upper aerodigestive tract in real-time[5]
 - The VFSS also permits detection of the presence and timing of aspiration[5]
 - With the VFSS, clinicians can observe the effects of various bolus volumes, bolus textures, and compensatory strategies on swallowing physiology[5]
- Participants attended a single visit and underwent the following procedures (**Figure 2**):
 - Baseline VFSS (×11 bolus trials)
 - Single dose ROF 50 mg placed on the median sulcus of the tongue
 - Post-ROF VFSS (×11 bolus trials)

Figure 2. Experimental Timeline of Procedures During Single Visit. Participants Underwent 2 Identical Vfsss Before and After Administration of ROF


1 Pre-ROFVFSS
2 ROF Administration
3 minutes
3 Post-ROFVFSS

RESULTS

Study 162020

- All (32) subjects received at least one dose of the study medication and comprised the safety population
- Thirty (30) individuals completed all 5 treatment periods
- Baseline characteristics are shown in **Table 1**

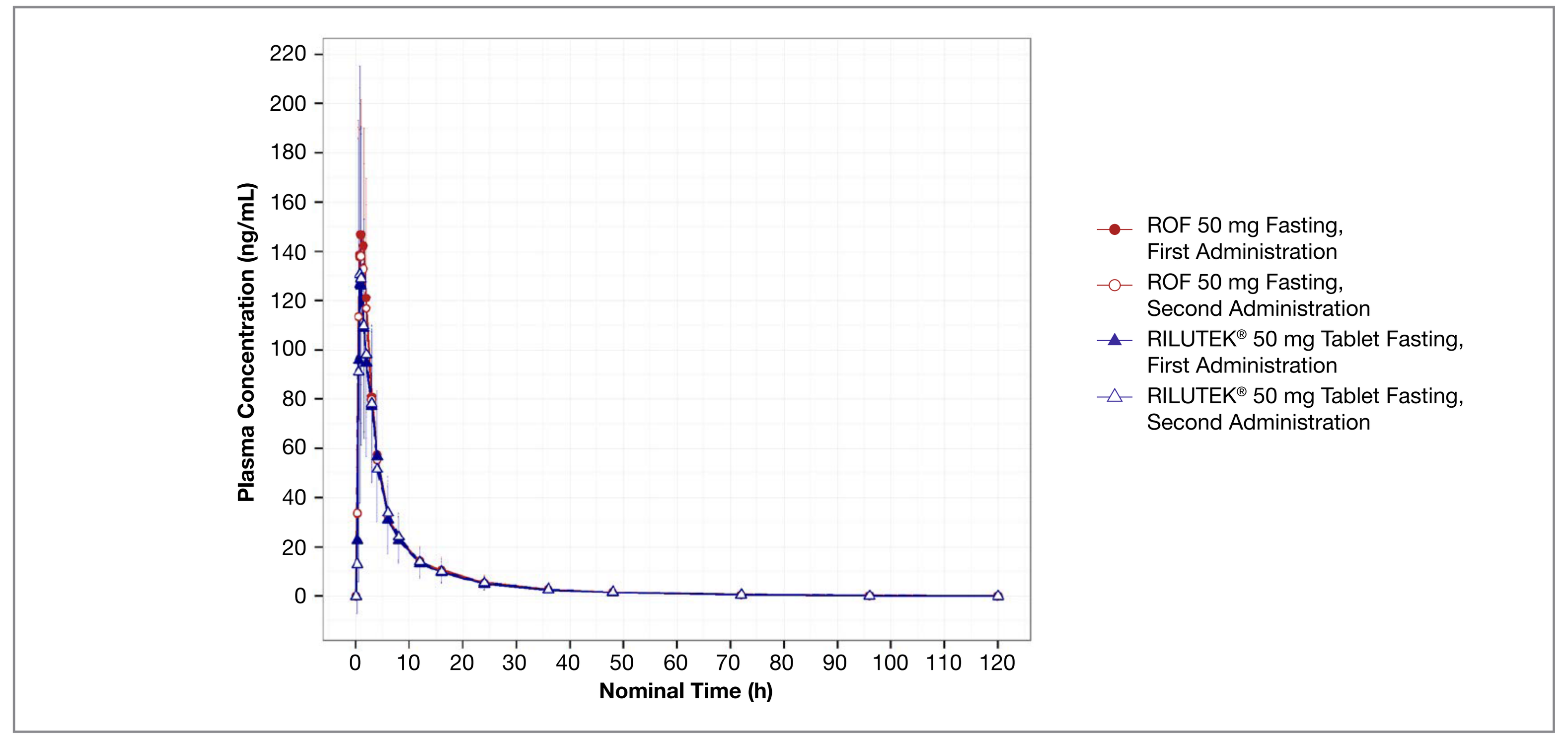
Table 1. Baseline Characteristics in Study 162020

Baseline Characteristic	Subjects N = 32
Age, mean years (SD)	37.6 (10.8)
Female, n (%)	16 (50)
Body weight, mean kg (SD)	71.6 (16.9)
Ethnicity, n (%)	
Not Hispanic or Latino	2 (6.3)
Hispanic or Latino	30 (93.8)
Height, mean cm (SD)	169.18 (7.98)
Weight, mean kg (SD)	72.40 (9.53)
Body mass index, mean kg/m² (SD)	25.28 (2.78)

Bioequivalence Under Fasting Conditions

- Overall, the PK parameters for ROF and Rilutek were comparable between treatments and administrations when administered under fasting conditions (**Figure 3**)

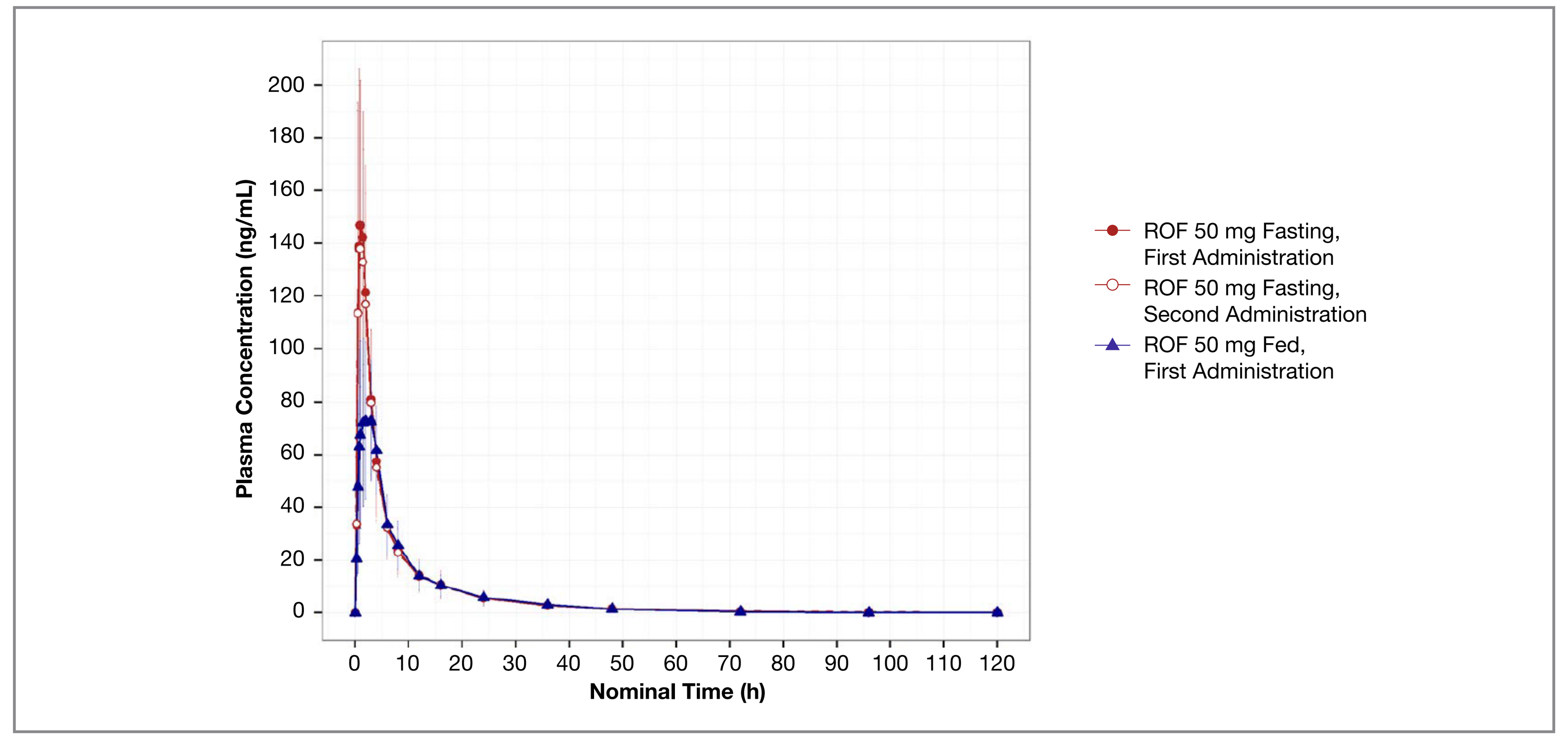
Figure 3. Riluzole Plasma Concentration (Treatments A and B)


Plasma Concentration (ng/mL)
Nominal Time (h)
ROF 50 mg Fasting, First Administration
ROF 50 mg Fasting, Second Administration
RILUTEK® 50 mg Tablet Fasting, First Administration
RILUTEK® 50 mg Tablet Fasting, Second Administration

Food Effect for the ROF

- Administration of ROF with food resulted in a 15% reduction in area under the curve and a 45% reduction in maximum serum concentration (**Figure 4**)

Figure 4. Riluzole Plasma Concentration (Treatments A and C)


Plasma Concentration (ng/mL)
Nominal Time (h)
ROF 50 mg Fasting, First Administration
ROF 50 mg Fasting, Second Administration
ROF 50 mg Fed, First Administration

Safety Data

- A total of 44 treatment-emergent adverse events (TEAEs) were reported in the study; all were mild in severity (**Table 2**)
- No serious AEs were observed, and no subjects discontinued due to AEs
- Among 32 subjects who received at least 1 dose of the ROF in Study 162020, 17 events in 14 subjects were characterized as oral or pharyngeal hypoesthesia
 - 17 events
- 16 events were coded as oral hypoesthesia and 1 event as pharyngeal hypoesthesia
- 14 occurred following ROF in the fasted condition
- 3 occurred following ROF in the fed condition

Table 2. Treatment-Emergent Adverse Events Reported in >1 Subject in Any Treatment Group

MedDRA® System Organ Class MedDRA® Preferred Term	Statistic	Treatment A ROF 50 mg Fasting Conditions (N=32)	Treatment B Rilutek® 50 mg tablet Fasting Conditions (N=31)	Treatment C ROF 50 mg Fed Conditions (N=30)	Overall (N=32)
Number of Treatment-Emergent Adverse Events (TEAEs)	E	23	8	13	44
Number of subjects with TEAEs	n (%)	15 (46.9)	7 (22.6)	8 (26.7)	19 (59.4)
Gastrointestinal disorders					
Hypoesthesia oral	n (%) E	12 (37.5) 13	0	3 (10.0) 3	14 (43.8) 16
Nervous system disorders					
Somnolence	n (%) E	5 (15.6) 6	5 (16.1) 5	3 (10.0) 3	9 (28.1) 14

E: Number of treatment-emergent adverse events.

Study 17MOIR-0012
Baseline Characteristics and PAS Scores

- The swallowing study included 9 patients with ALS who had a mean age of 61.6 years (**Table 3**)
- Baseline disease characteristics and PAS pre- and post-scores are shown in **Table 4**

Table 3. Study 17MOIR-0012 Participant Demographics

Age	Gender	ALSFRS-R	BMI
Mean (SD): 61.6 (12.0) (range 37 – 77)	66% Female 34% Male	Median: 38 (4.9) (range: 27 – 44)	Mean (SD): 27.7 (5.6) (range: 19.4, 35.9)

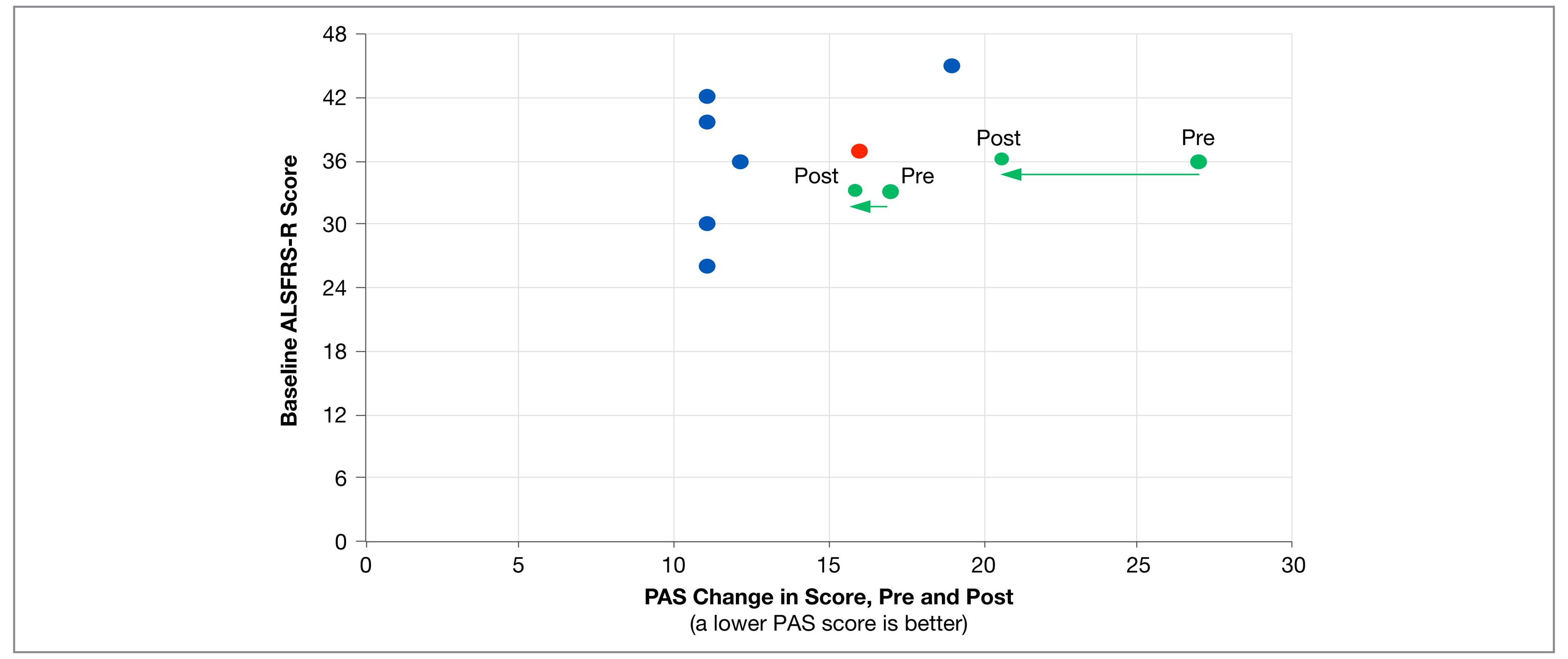
Table 4. Study 17MOIR-0012 Participant Disease Characteristics and PAS Scores

Patient	Age	Sex	BMI	ALSFRS-R	ALSFRS-R bulbar score	EAT-10	PAS pre-dose score	PAS post-dose score
1	68	M	23.8	40	11	0	11	11
2	60	F	30	36	12	4	27	21
3	68	M	30	36	12	4	12	12
4	50	F	19.4	33	12	0	17	16
5	67	M	22.3	45	12	1	19	19
6	77	F	35.9	37	12	0	16	16
7	69	F	33.7	30	9	2	11	11
8	37	F	31	42	10	4	11	11
9	58	F	29.5	26	12	0	11	11

EAT-10, Eating Assessment Tool-10

- In the swallowing study, 55.6% of patients were able to swallow safely by PAS standards both pre- and post-dose
- Very little numerical or categorical change was observed following the dose of ROF
- No evidence of deterioration of swallowing function was observed post-dose
- There was no correlation between baseline ALSFRS-R score and changes in PAS scores (**Figure 5**)
- Similar findings were obtained for ALSFRS-R bulbar domain scores vs PAS scores and for EAT-10 scores vs PAS scores

Figure 5. Swallowing Study PAS Scores vs Baseline ALSFRS-R Score


Baseline ALSFRS-R Score
PAS Change in Score, Pre and Post
(a lower PAS score is better)
Post
Pre
Post
Pre

In the figure, green circles represent patients who improved in their PAS scores from pre- to post-dose, the red circle represents a patient who had worsening of her PAS worst score from pre-dose to post-dose.

DISCUSSION

- The ROF was bioequivalent to Rilutek tablets and was well tolerated in healthy adults
- A single dose of 50 mg ROF had no detrimental effect on swallowing, indicating that ROF represents an alternative administration route in patients with ALS who have dysphagia

REFERENCES

1. Hardiman O, Al-Chalabi A, Chio A, et al. Amyotrophic lateral sclerosis. *Nat Rev Dis Primers*. 2017;3:17071.
2. Rilutek® (riluzole) [package insert]. Cary, NC: Covis Pharmaceuticals, Inc.; April 2016.
3. EXSERVAN™ (riluzole) oral film [package insert]. Jersey City, NJ: Mitsubishi Tanabe Pharma America, Inc.; April 2021.
4. Costa MM. Videofluoroscopy: the gold standard exam for studying swallowing and its dysfunction. *Arq Gastroenterol*. 2010;47(4):327-328.
5. Martin-Harris B, Jones B. The videofluorographic swallowing study. *Phys Med Rehabil Clin N Am*. 2008;19(4):769-785, viii.

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